

Supporting information

Phase transformation of GeO₂ glass to nanocrystals under ambient condition

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Materials and methods

Sample preparation

GeO₂ glass was prepared by using the conventional melt-quenching method, with commercial GeO₂ (99.99%, Yunan Germanium Inc.). 100 g GeO₂ powder was melted in an alumina crucible at 1500°C for 2 h in air. The homogeneous melts were poured onto a preheated stainless steel mold and annealed at 600°C for 2 h, followed by cooling down naturally to room temperature. Part of the as-obtained GeO₂ glass was ground into powder and finally sieved through 300 meshes.

Characterization

XRD patterns of the pure GeO₂ glass powder (0.5 g) and the hydrated glass powder in the presence of small amount of deionized water (0.2 ml) for different time durations (i.e., 5, 15, 30 and 60 min) were recorded by using a D/Max-2500 diffractometer with nickel-filtered Cu K α radiation at a tube voltage of 40 kV. The scan speed is set to be 6 °/min over 10° - 70°. The specimen for XRD testing was filled in the groove of a glass slide. In order to avoid the evaporation of moisture, a polyethylene film was employed to wrap the glass slide. These specimens were also observed using scanning electron microscopy (SEM, JSM-6700F).

FTIR transmittance spectra of GeO₂ glass before and after the hydration of GeO₂ glass powder were collected by using a Nicolet 5700 spectrometer. The FTIR samples were prepared through the standard KBr pellet method. In particular, the as-sieved glass powder, 0.5 g glass powder blended with 0.2 ml deionized water after holding for 5, 15, 30 and 60 min, and 99.99% purity of dried hexagonal GeO₂ crystal were chosen to be examined.

Raman spectroscopy was also performed to further clarify the microstructure of GeO₂ glass (0.5 g) before and after the hydration in 0.2 ml deionized water for time durations of 5, 15, 30 and 60 min. The dried hexagonal GeO₂ crystal was also examined as reference.

The change in pH value of the mixture of GeO₂ glass powder and water during the hydration as a function of time was monitored by using a Hanna HI3512 model pH meter. 0.5 g GeO₂ glass powder was added into 50 ml deionized water forming a suspension. During the test, this suspension was continuously stirred with a magnet. The pH value was recorded at the interval of 30 s.

A TEM liquid cell holder attached to a field emission JEOL-TEM2100F (JEOL Ltd.) operated at 200 kV (Poseidon Select, Protochips) was used to carry out the in-situ observation of the microstructural evolution during phase transformation of GeO₂ glass in water. The fluid cell consists of two E-chips, with sizes of 2 mm × 2 mm and 6 mm × 4.5 mm, respectively. The two E-chips are packaged together and sealed with an O-ring. The small E-chip has a 50 nm thick transparent window with the size of 50 μm × 550 μm and has a deposited oxide layer on the internal face to create a rectangular flow channel between the assembled cells (Figure S4). This allows liquid from the inlet to flow through the liquid cell from one side to the other across the windows. This imaging chamber is assembled within the tip of the specimen holder that is completely isolated from the vacuum system when inserted into the TEM chamber. Movie files were constructed using OpenShot Video Editor software (OpenShot Studios, LLC) based on the sequential screenshot images taken by Digital Micrograph (Gatan Inc.)

To prepare the TEM samples, GeO₂ glass powder (300#, 0.5 g) was mixed with absolute alcohol (50 ml) and further ground with agate mortar forming a suspension. After holding for 10 min, the suspension was dropped on the bottom membrane of the chip and then sealed after the complete evaporation of alcohol.

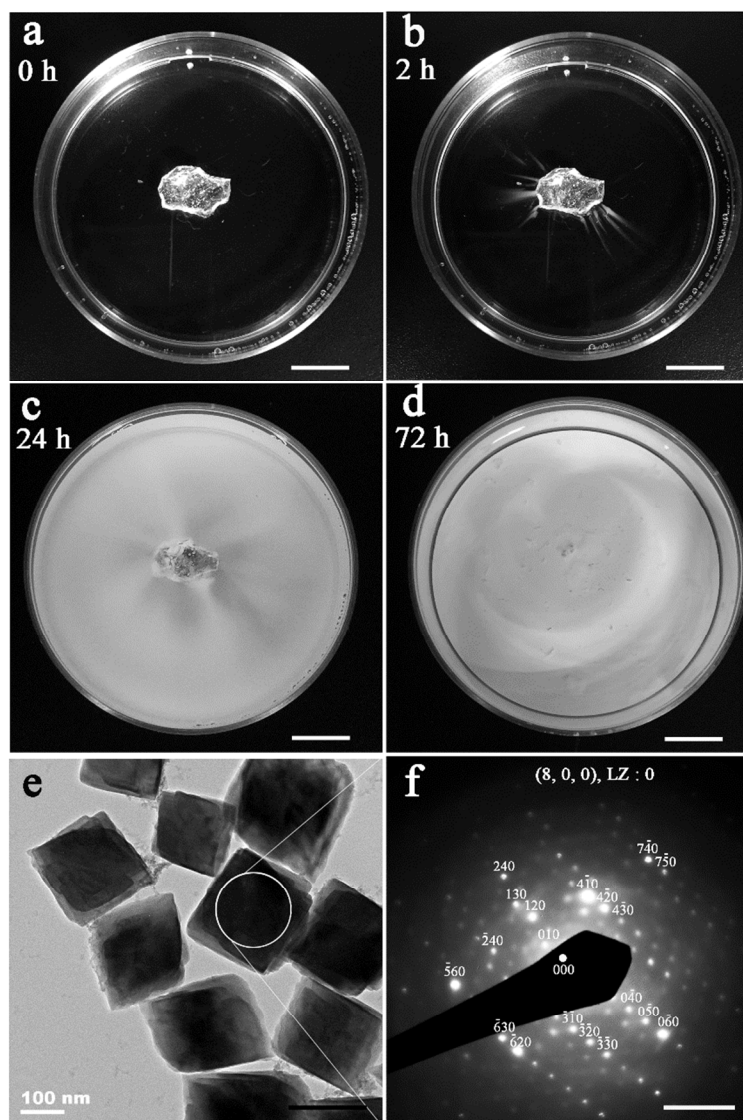


Figure S1. A piece of GeO₂ glass completely converts into crystal particles in large amount of water. a-d) Optical photos showing phase transformation process at room temperature through hydrolysis without agitation. a) It starts with a piece of GeO₂ glass (1.105 g) that is immersed in water (20 ml), b) which is partly dissolved after 2 h. c) After 24 h, the GeO₂ piece is further diminished, d) which completely turns to very small dots in 72 h. e) Square shaped particles were observed when the dots were viewed under TEM. f) Selected-area diffraction pattern (taken from the marked area in e)), indicating the lattice perfection of the particles. Scale bars are 5 mm in a-d), 100 nm in e) and 10 nm⁻¹ in f).

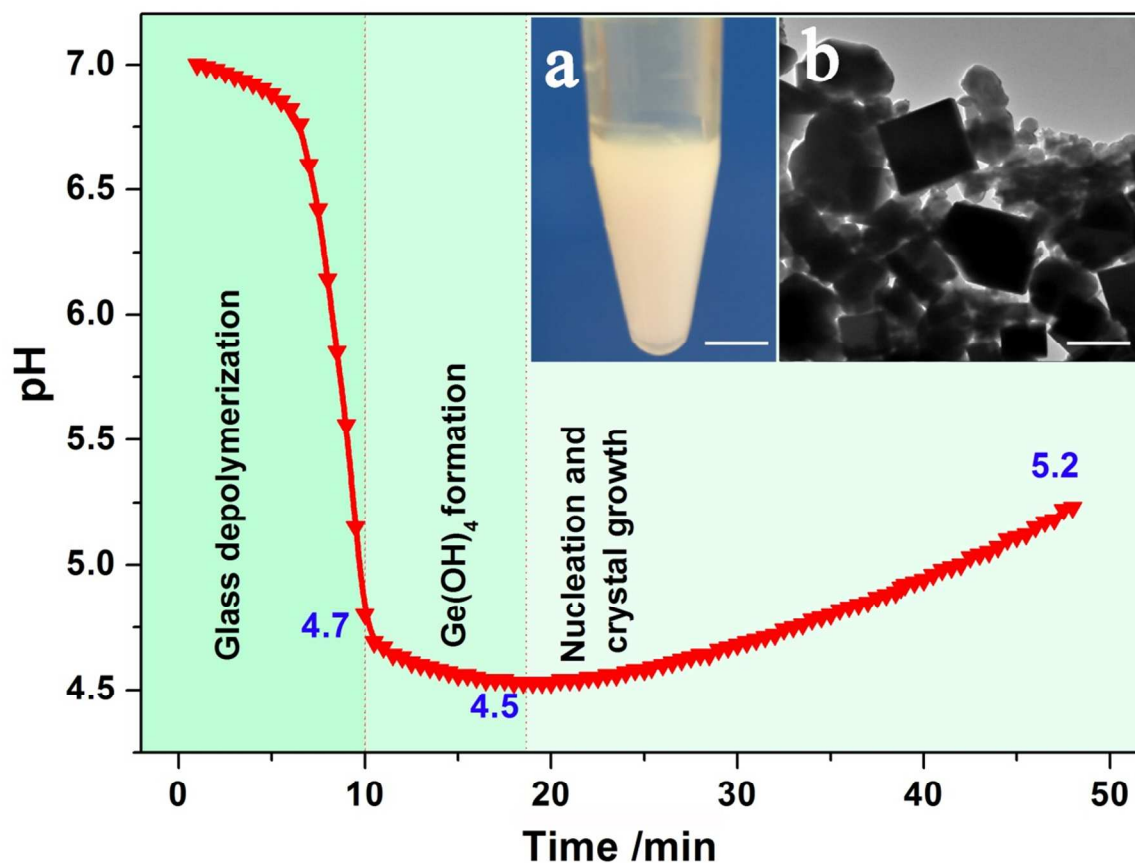


Figure S2. Variation in pH value of the aqueous suspension as a function of reaction time. According to the change in pH of the suspension, the reaction process can be roughly divided into three regions: depolymerization of GeO₂ glass network, formation of germanate acid (Ge(OH)₄) and precipitation of nanocrystal of GeO₂. a) Optical photo of the suspension (about 1 ml) after stirring for 50 min. b) Corresponding SEM image of the particles derived from the suspension with regular cubic shape. Scale bars are 5 mm in a) and 200 nm in b).

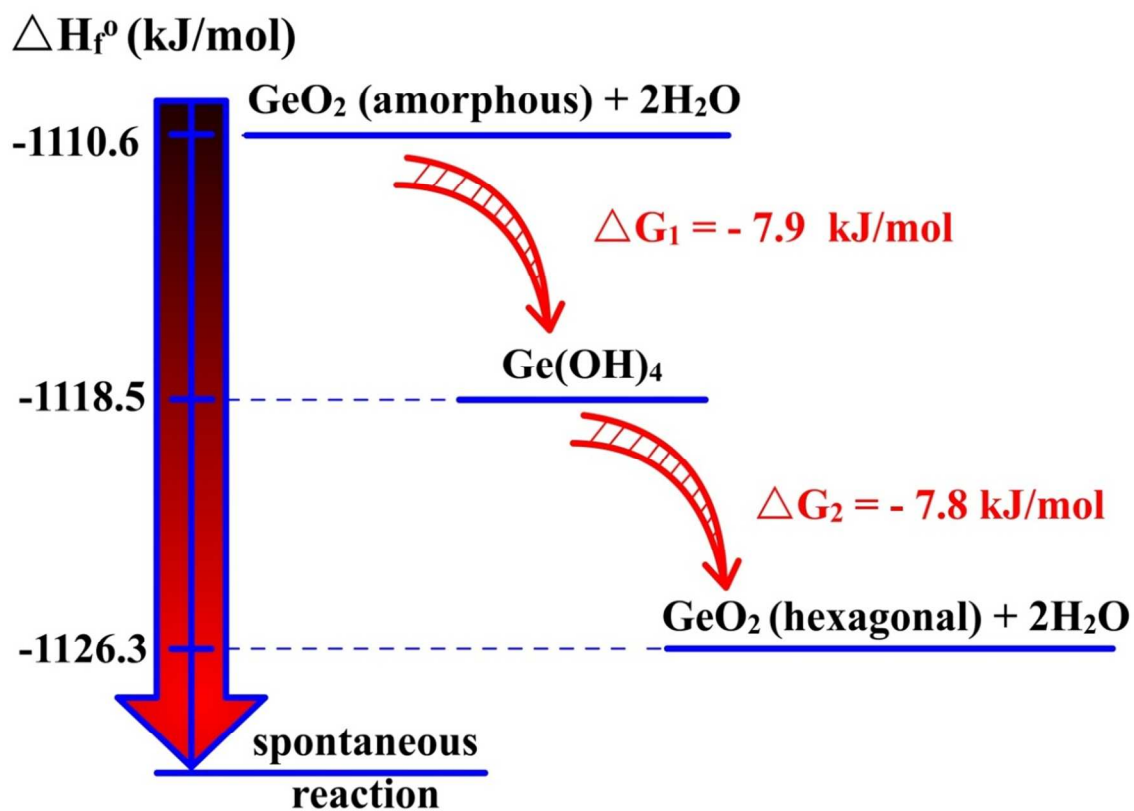


Figure S3. Diagram of Gibbs free energies for the overall reaction during the phase transformation of GeO_2 glass to nanocrystal in the presence of water.

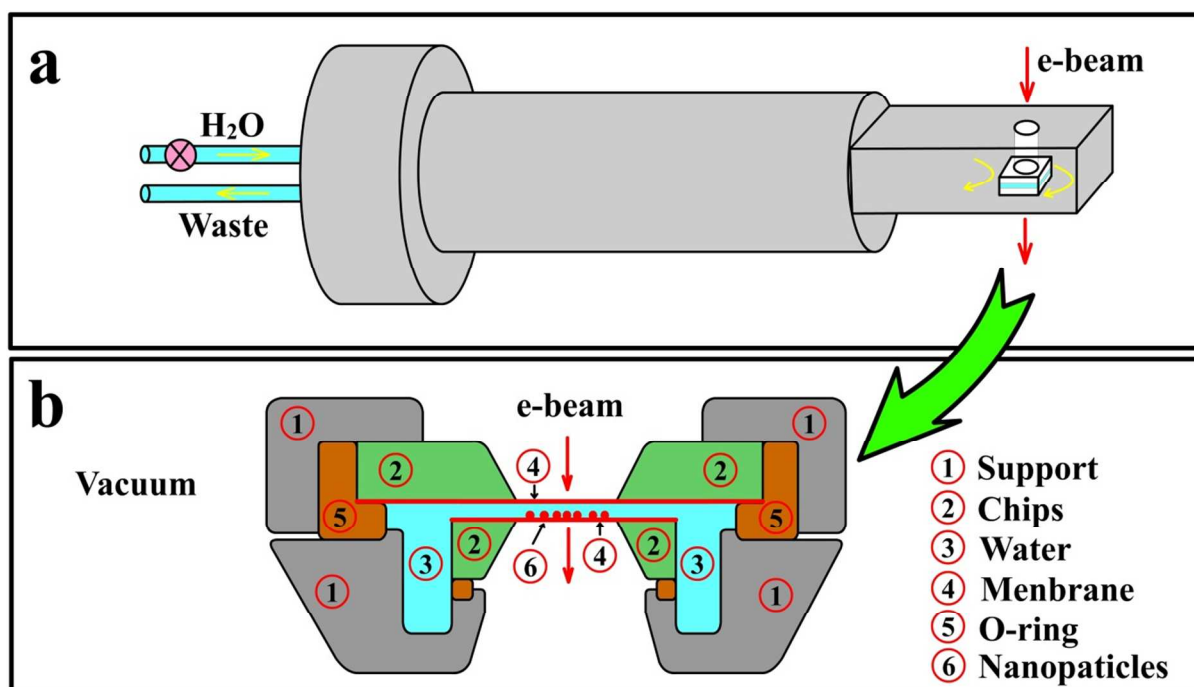


Figure S4. Schematic diagrams of the in-situ TEM holder and liquid cell. a) Schematic diagrams of the in-situ TEM holder. b) Diagrammatic sketch of the liquid cell in the in-situ TEM holder. The fluid cell consists of two E-chips, with sizes of 2 mm × 2 mm and 6 mm × 4.5 mm, respectively. The two E-chips are packaged together and sealed with an O-ring. The small E-chip has a 50 nm thick transparent window with the size of 50 μm × 550 μm and has a deposited oxide layer on the internal face to create a rectangular flow channel between the assembled cells. This allows liquid from the inlet to flow through the liquid cell from one side to the other across the windows. This imaging chamber is assembled within the tip of the specimen holder that is completely isolated from the vacuum system when inserted into the TEM chamber.

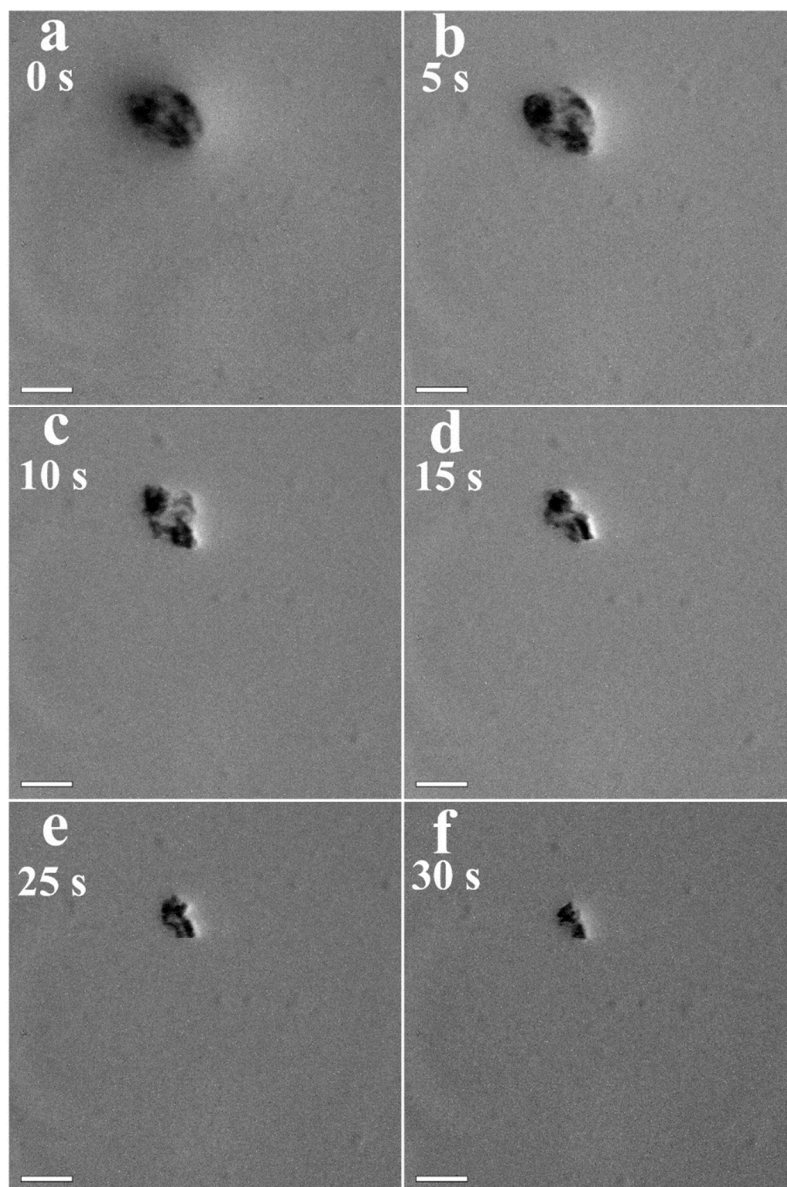


Figure S5. In-situ TEM observation of hydrolysis of the GeO₂ glass particles in deionized water (original TEM images of Figure 2.) The in-situ TEM images clearly show the hydrolysis process of the GeO₂ glass particles in water with different duration times: (a) 0 s, (b) 5 s, (c) 10 s, (d) 15 s, (e) 25 s and (f) 30 s. Scale bars are 200 nm.

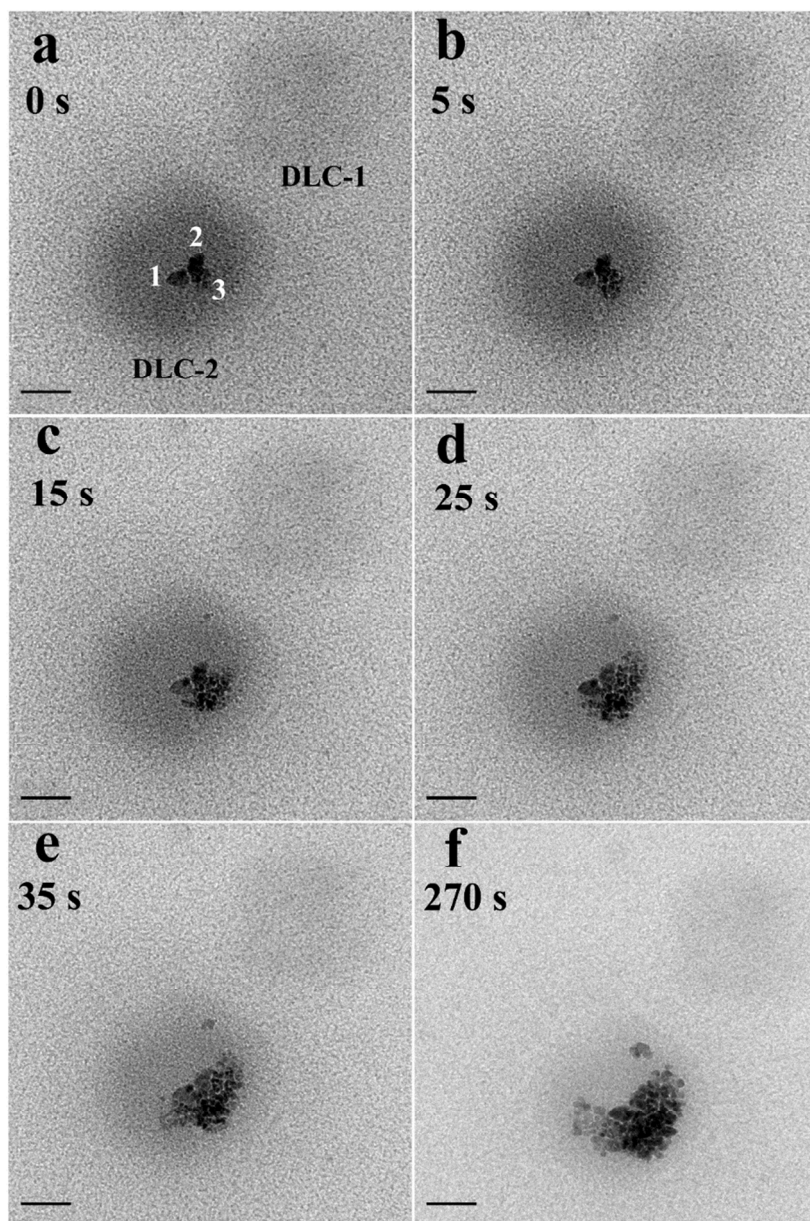


Figure S6. In-situ TEM observation of nucleation and crystallization of hexagonal GeO_2 nanocrystals in deionized water (Original images of Figure 3). a) Two DLCs (DLC-1 and DLC-2) and several incomplete dissolved glass particles (labeled as 1, 2 and 3) that can be observed in the initial TEM image. b-f) The nucleation and crystallization process of the GeO_2 nanocrystals accompanied by the rapid consumption of clusters can be clearly observed. Scale bars are 50 nm.

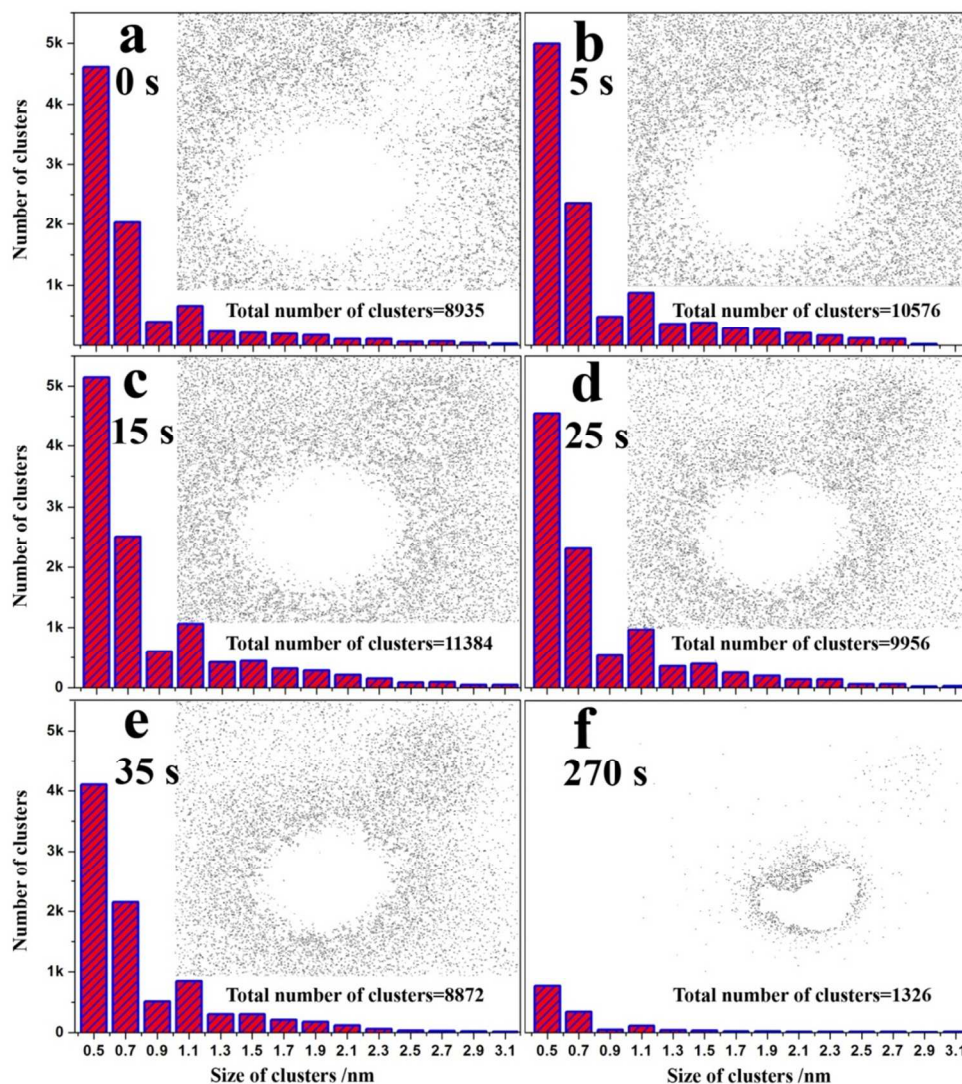


Figure S7. Statistical plots of cluster size and distribution derived from selected sequential in-situ TEM images: (a) 0 s, (b) 5 s, (c) 15 s, (d) 25 s, (e) 35 s and (f) 270 s. The data are derived from the automatic analysis results by using the software ImageJ R1.51 (“analyze particles module”) with fixed analytical parameters. The clusters in the DLC-2 are not counted by using the software, because they are too dense to identify.

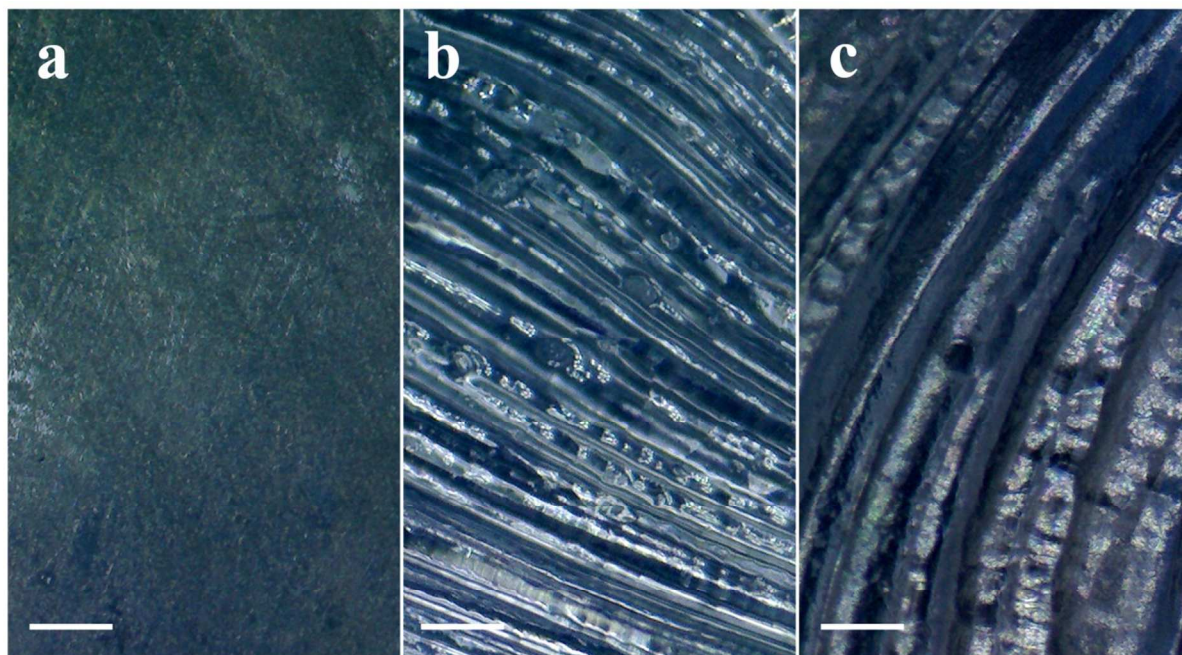


Figure S8. Optical microscopy photos showing the surface morphologies of bulk GeO_2 glass before and after reaction with water. a) Original flat surface of GeO_2 glass block obtained by using diamond cutting discs. b) New surface of the glass which was immersed in deionized water for 8 h where the surface crystalline layer is removed by ethanol rinsing. c) High magnification of the photo b), oscillatory spatial patterns can be observed on the newly formed surface due to the reaction with deionized water. Scale bars are 50 μm in a) and b), and 25 μm in c).

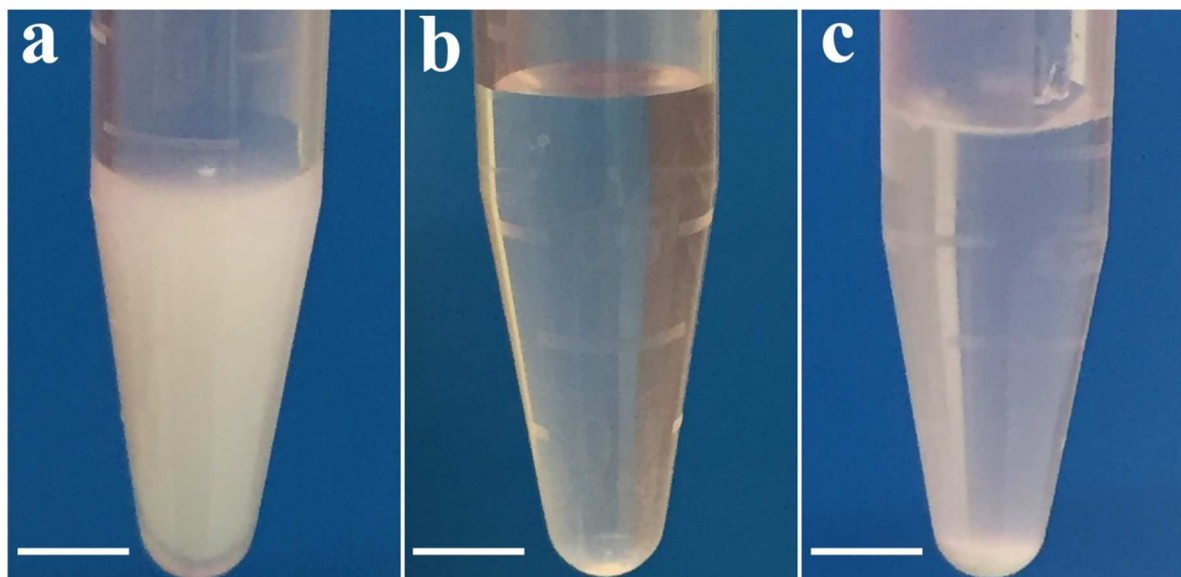


Figure S9. Optical photos showing the effect of pH value on stability of the GeO_2 glass suspension in water. 0.5 g GeO_2 glass powder was added into 50 ml deionized water and stirred with a magnet to form a suspension, from which three samples were prepared with about 1 ml each. a) Original mixture (with weak acidic, $\text{pH} \approx 5$). b) Transparent solution upon addition of ammonia ($\text{pH} \approx 10$). c) Precipitation observed immediately after addition of HCl ($\text{pH} \approx 1$).

Video S1

A movie file showing the hydrolysis process of the GeO_2 glass particles in deionized water. The size of the particles is seen to gradually decrease and the background appears darker and darker due to the diffusion of the hydrolyzate. The scale bar is 200 nm.

Video S2

A movie file showing the nucleation and crystallization process of the GeO_2 crystals in aqueous suspension. Two relatively dark areas of spherical shape are the dense liquid clusters formed due to the aggregation of $\text{Ge}(\text{OH})_4$ clusters in the solution. The scale bar is 50 nm.